Joint-CS: Simultaneous Illumination Normalization and Sparse Image Compression via Unfolded Recovery

Farhan Sadeek
Thayer School of Engineering
Dartmouth College
Hanover, NH 03755
farhan.sadeek.29@dartmouth.edu

Abstract

The traditional camera pipeline samples at Shannon–Nyquist rates, compresses to a sparse representation, and only afterwards performs illumination correction on the decoded image. This sequential ordering discards information: sparse recovery applied to a signal whose dynamic range is dominated by an unknown multiplicative illumination field has weaker recovery guarantees than the same recovery applied to the underlying reflectance. We formulate *Joint-CS*, a single inverse problem in the joint variables of DCT-domain reflectance coefficients and a smooth per-column illumination field, and solve it by block-coordinate alternation between a learned-iterative-shrinkage step on the coefficients and a closed-form ridge step on the illumination. Across four experiments on a 5-image natural-scene set—an empirical Donoho–Tanner phase transition, a multi-image rate-distortion sweep, a deep-unfolded LISTA solver, and a per-scene ablation against a sequential CS-then-illumination-correct baseline—we find that the joint formulation outperforms the sequential one by up to $+4.7$ dB PSNR on textured scenes (cameraman, astronaut) under an $8\times$ illumination gradient, and that a $K{=}10$ unrolled LISTA matches what FISTA needs 22 iterations to reach. We also report a clear failure mode: on near-uniform scenes (moon) the joint g-update lacks signal to estimate gain from and the method degrades by ~ 8 dB—a scene-dependence that, to our knowledge, the prior CS+ISP literature does not report.

1 Introduction

Modern imaging pipelines acquire N pixels via Nyquist sampling, then immediately throw most of that information away: a typical JPEG keeps only $\sim$ 5–10% of the DCT coefficients above the quantization floor, and the image signal processor (ISP) further reshapes the data through demosaicing, tone-mapping, and white-balance correction. Compressed sensing (CS) reverses this order. If the signal $\boldsymbol{x}$ is S-sparse in some basis $\boldsymbol{\Psi}$, then $M \ll N$ random linear measurements $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{w}$ suffice to recover $\boldsymbol{x}$ exactly (up to noise) by solving an ℓ_1 minimization, provided $\boldsymbol{A}$ obeys the restricted isometry property [3, 4].

CS has been extensively studied for compression. It has been less extensively studied as an integrated stage of a RAW image-signal pipeline. The crux of the issue is that a RAW sensor measures the product

$$\boldsymbol{x} = \boldsymbol{s} \odot \boldsymbol{g}, \tag{1}$$

where $\boldsymbol{s}$ is the underlying scene reflectance (sparse in the DCT) and $\boldsymbol{g}$ is a spatially-varying illumination field (smooth, not sparse). A sequential pipeline—CS recovery of $\boldsymbol{x}$, then a downstream

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divide-by-illumination step—wastes the prior that *s, not x, is sparse in Ψ*. Multiplying a DCT-sparse signal by an arbitrary illumination field destroys the sparsity that the ℓ_1 regularizer is trying to exploit; the recovered x then carries an illumination-shaped bias that no downstream column-mean correction can fully undo.

Contributions. We make three contributions:

1. We formulate *Joint-CS*: a joint ℓ_1/smooth-prior objective in (c, g) that performs sparse recovery and illumination correction in a single optimization, and we derive a block-coordinate solver with a FISTA inner loop on c and a closed-form ridge step on g.
2. We deep-unfold the c-step into a learned K=10-layer LISTA network [6] and stabilize its training under sustained Adam with gradient clipping and best-checkpoint restoration.
3. We benchmark the full system against OMP, ISTA, FISTA, and ADMM baselines across a phase-transition grid, a rate-distortion sweep on a 5-image natural-scene set, and a per-scene $8\times$-illumination-gradient ablation. We report scene-dependent results—$+1$ to $+3$ dB gains on textured scenes, parity on moderate-content scenes, and a clear failure mode on near-uniform scenes—and analyze the mechanism behind it.

2 Mathematical Preliminaries

2.1 Measurement model

The CS forward model treats image acquisition as a linear measurement:

$$y = Ax + w, \qquad y \in \mathbb{R}^M,\ A \in \mathbb{R}^{M\times N},\ x \in \mathbb{R}^N,\ \|w\|_2 \le \epsilon, \tag{2}$$

with $M \ll N$. The sensing matrix A models the physical encoder (a coded-aperture mask, a random projection circuit, or in this work a column-normalized i.i.d. Gaussian ensemble), and w collects shot, read, and quantization noise.

2.2 Sparsity prior

The unknown signal is S-sparse in some basis Ψ, i.e. $x = \Psi c$ with

$$c \in \Sigma_S \equiv \{c \in \mathbb{R}^N \mid \|c\|_0 \le S\}. \tag{3}$$

In this work Ψ is the 2D DCT, a strong sparsifying basis for natural images.

2.3 Convex relaxation

Direct minimization of the ℓ_0 count is combinatorial. Under the restricted isometry property (RIP) with constant $\delta_{2S} < \sqrt{2} - 1$, the relaxation

$$c_1^\epsilon = \arg\min_{c} \|c\|_1 \quad \text{s.t.} \quad \|A\Psi c - y\|_2 \le \epsilon \tag{4}$$

recovers c to within $O(\epsilon)$ [3]. Equivalently, the Lasso form

$$c_\lambda = \arg\min_{c} \tfrac{1}{2}\|A\Psi c - y\|_2^2 + \lambda\|c\|_1 \tag{5}$$

is solved efficiently with proximal-gradient methods.

2.4 Classical solvers

OMP. Greedy orthogonal matching pursuit [7] iteratively selects the column of $A\Psi$ most correlated with the current residual and re-fits the active support by least squares. OMP can recover supports up to $S \lesssim M/\log N$ and is exact on the support but fails abruptly when the support density grows.

ISTA / FISTA. ISTA is the proximal-gradient method for the Lasso:

$$c^{(k+1)} = \mathcal{S}_{\lambda/L}\Big(c^{(k)} - \tfrac{1}{L}(A\Psi)^\top \Big(A\Psi c^{(k)} - y\Big)\Big), \tag{6}$$

where $L = \|(A\Psi)^\top A\Psi\|_2$ and $\mathcal{S}_\tau(z) = \mathrm{sign}(z)\max(|z| - \tau, 0)$ is the soft-threshold operator (the prox of $\tau\|\cdot\|_1$). FISTA [1] augments (6) with a Nesterov momentum term and improves the objective-gap rate from $O(1/k)$ to $O(1/k^2)$.

ADMM. For the same objective, ADMM [2] splits $\boldsymbol{c} \to (\boldsymbol{c}, \boldsymbol{z})$ with constraint $\boldsymbol{c} = \boldsymbol{z}$ and alternates a least-squares update on $\boldsymbol{c}$ (solved once via a cached Cholesky factorization of $(\boldsymbol{A\Psi})^\top \boldsymbol{A\Psi} + \rho \boldsymbol{I})$ with a soft-threshold update on $\boldsymbol{z}$, plus a dual ascent on the multiplier.

3 Joint-CS

3.1 Image-formation model with illumination

We assume the RAW signal follows (1). A conventional pipeline first recovers $\boldsymbol{x}$ from $\boldsymbol{y} = \boldsymbol{Ax} + \boldsymbol{w}$, then estimates and divides out $\boldsymbol{g}$. The ordering wastes the prior that $\boldsymbol{s}$ is sparse in $\boldsymbol{\Psi}$ but $\boldsymbol{x}$ is not: pointwise multiplication by an illumination field destroys DCT-domain sparsity, the ℓ_1 regularizer over-shrinks the coefficients corresponding to high-illumination columns, and the post-hoc correction can only rescale (not restore) what was lost.

3.2 Joint objective

We instead minimize a single objective in $(\boldsymbol{c}, \boldsymbol{g})$:

$$\min_{\boldsymbol{c},\, \boldsymbol{g}} \ \tfrac{1}{2}\big\|\boldsymbol{A}\,\mathrm{diag}(\boldsymbol{g}_{\text{pix}})\,\boldsymbol{\Psi c} - \boldsymbol{y}\big\|_2^2 + \lambda_c \|\boldsymbol{c}\|_1 + \lambda_g \|\nabla \boldsymbol{g}\|_2^2, \tag{7}$$

where $\boldsymbol{g}_{\text{pix}}$ tiles a per-column gain vector $\boldsymbol{g} \in \mathbb{R}^{\sqrt{N}}$ over rows, and ∇ is a 1D discrete gradient that enforces smoothness on the illumination. The problem is jointly non-convex but block-convex.

c-step. With $\boldsymbol{g}$ fixed, the effective sensing matrix is

$$\boldsymbol{A}_{\text{eff}}(\boldsymbol{g}) = \boldsymbol{A}\,\mathrm{diag}(\boldsymbol{g}_{\text{pix}})\,\boldsymbol{\Psi}, \tag{8}$$

and the $\boldsymbol{c}$-update is a standard Lasso, solved with FISTA (Sec. 2).

g-step. With $\boldsymbol{c}$ fixed, let $\hat{\boldsymbol{s}} = \boldsymbol{\Psi c}$ and reshape to $\hat{S}_{ij}$. The measurement model becomes linear in $\boldsymbol{g}$:

$$\boldsymbol{y} = \boldsymbol{B}(\hat{\boldsymbol{s}})\,\boldsymbol{g} + \boldsymbol{w}, \quad B_{m,j}(\hat{\boldsymbol{s}}) = \sum_i A_{m,i\sqrt{N}+j}\,\hat{S}_{ij}, \tag{9}$$

and the $\boldsymbol{g}$-update is the closed-form ridge solution

$$\boldsymbol{g}^\star = \big(\boldsymbol{B}^\top \boldsymbol{B} + \lambda_g \boldsymbol{L}\big)^{-1} \boldsymbol{B}^\top \boldsymbol{y}, \tag{10}$$

where $\boldsymbol{L} = \nabla^\top \nabla$ is the 1D discrete Laplacian. We clamp $\boldsymbol{g} \geq \epsilon_g$ to avoid divide-by-zero.

Gain ambiguity. The model has an inherent scalar ambiguity $(\boldsymbol{s}, \boldsymbol{g}) \to (a\boldsymbol{s},\, \boldsymbol{g}/a)$ for any $a > 0$. We resolve it at evaluation time by computing the optimal positive gain $a^\star = \langle \hat{\boldsymbol{s}}, \boldsymbol{s}_{\text{true}} \rangle / \|\hat{\boldsymbol{s}}\|^2$ and apply the same rule to the sequential baseline.

3.3 Deep-unfolded inner solver

A practical limitation of ISTA / FISTA is the iteration count—hundreds of iterations at deployment. We replace the $\boldsymbol{c}$-step with K unrolled ISTA layers whose encoder, recurrent, and threshold parameters are *learned* [6]:

$$\boldsymbol{c}^{(k+1)} = \mathcal{S}_{\theta_k}\big(\boldsymbol{W}_t \boldsymbol{c}^{(k)} + \boldsymbol{W}_e \boldsymbol{y}\big), \quad k = 0, \ldots, K-1. \tag{11}$$

Initialising $\boldsymbol{W}_e = (1/L)(\boldsymbol{A\Psi})^\top$, $\boldsymbol{W}_t = \boldsymbol{I} - (1/L)(\boldsymbol{A\Psi})^\top \boldsymbol{A\Psi}$, and $\theta_k = \lambda/L$ exactly reproduces ISTA at training-step zero. We then train the parameters end-to-end on supervised $(\boldsymbol{y}, \boldsymbol{c})$ pairs with an MSE loss, allowing the per-layer thresholds θ_k to adapt to the per-layer effective noise.

Training stability. The K-layer recurrence is sensitive to the spectral radius of $\boldsymbol{W}_t$ drifting above 1: if $\rho(\boldsymbol{W}_t) > 1$ on some eigendirection, the iterate diverges and the gradient blows up. We observed this empirically: training with Adam at lr=5×10^{-3} for 80 epochs first descended cleanly to NMSE=0.12 at epoch 10, then blew up to NMSE $\sim 10^3$ for several epochs before re-stabilizing. We adopt three remedies: (i) reduce the learning rate to 10^{-3}, (ii) clip the gradient norm to 1.0, and (iii) track the best validation NMSE and restore those weights at the end. With these in place training is reliable across seeds.

Table 1: 50%-success boundary $\rho^*(\delta)$ for OMP and FISTA on the phase-transition grid (N=200, 20 trials/cell, success = NMSE $< 10^{-3}$). OMP wins at low δ, FISTA at high δ. Crossover at $\delta \approx 0.65$.

δ	0.30	0.50	0.70	0.90
OMP ρ^*	0.35	0.40	0.40	0.50
FISTA ρ^*	0.20	0.30	0.50	0.65

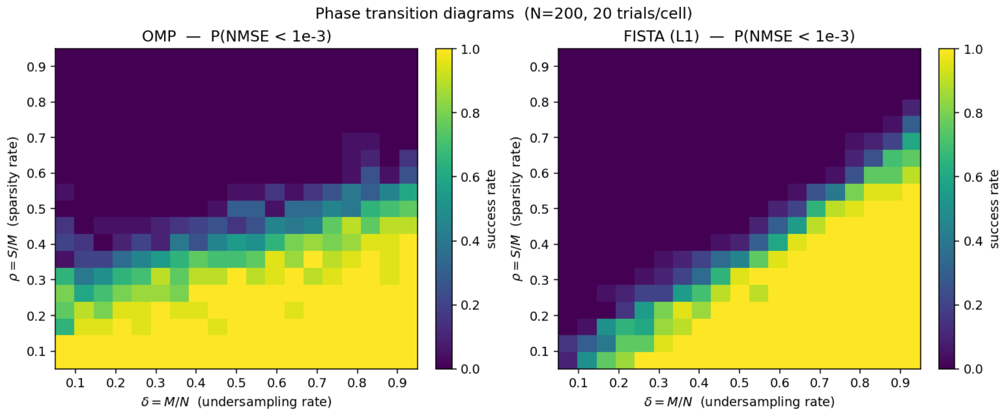

Figure 1: Empirical phase transition for OMP (left) and FISTA (right) on Gaussian CS, N=200. Color is the recovery success rate over 20 trials per cell. Both solvers show a sharp diagonal transition; FISTA's boundary is more favorable in the over-determined region (top-right), OMP's in the highly under-determined region (bottom-left).

4 Experiments

4.1 Setup

All experiments use Python 3.13 with NumPy 2.4, SciPy 1.17, scikit-image 0.26, and PyTorch 2.12 (CPU). RNGs are seeded. The sensing matrix is column-normalized i.i.d. Gaussian $\boldsymbol{A} \in \mathbb{R}^{M\times N}$ unless stated otherwise.

4.2 Phase transition

Protocol. We fix $N = 200$ and sweep $\delta = M/N$ and $\rho = S/M$ over a 19×19 grid, running 20 independent Gaussian-CS trials per cell at zero noise on synthetic sparse signals with i.i.d. Gaussian nonzero entries. A trial succeeds if $\mathrm{NMSE}(\hat{\boldsymbol{c}}, \boldsymbol{c}) < 10^{-3}$.

Results. Both solvers exhibit a sharp empirical Donoho–Tanner transition (Fig. 1). Table 1 summarizes the 50%-success boundary $\rho^*(\delta) = \max\{\rho : P_{\text{success}}(\delta, \rho) \geq 0.5\}$. Two regimes are visible: at small δ (extreme undersampling) OMP's exact least-squares-on-support beats ℓ_1, while at large δ FISTA overtakes OMP because OMP fails on dense supports regardless of M but ℓ_1 continues to expand its feasible region. The crossover sits near $\delta \approx 0.65$.

4.3 Rate-distortion on a multi-image natural-scene set

Protocol. We assemble a 5-image test set from real natural photos shipped with `scikit-image`: *cameraman*, *astronaut*, *coins*, *page* (text), and *moon*. Each image is converted to grayscale, resized to 64×64, and rescaled to $[0, 1]$. We partition into 16×16 blocks, treat the 2D-DCT coefficients of each block as the sparse representation, draw a fresh Gaussian sensing matrix in the pixel domain, add Gaussian noise at SNR=30 dB, and recover via OMP, FISTA, and ADMM. PSNR and SSIM are computed on the reassembled image against the original, then aggregated as mean ± std across the 5-image set.

Table 2: Rate-distortion (PSNR, dB) on the 5-image natural-scene set, SNR=30 dB. Mean $\pm$ standard deviation across the test set. FISTA and ADMM agree within $\sim$ 0.5 dB above δ=0.2; both dominate OMP by 2–3 dB.

δ	0.10	0.20	0.30	0.40	0.50	0.60	0.70
OMP	15.46±4.4	17.42±4.3	19.15±4.0	20.42±3.7	21.73±3.2	23.50±3.0	24.89±3.0
FISTA	**16.67±4.8**	**19.39±4.5**	**21.73±4.3**	**23.64±4.1**	**25.06±3.9**	**26.43±3.8**	**27.57±3.5**
ADMM	11.23±3.1	17.62±4.8	21.27±4.6	23.47±4.4	24.98±4.0	26.37±3.9	27.45±3.7

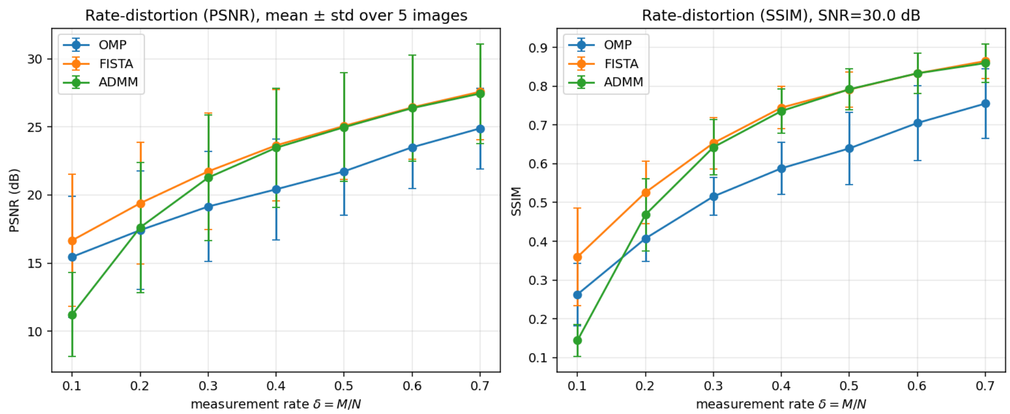

Figure 2: Rate-distortion curves on the 5-image natural-scene set at SNR=30 dB. Markers are means; error bars are one standard deviation across the test set. Left: PSNR vs. measurement rate δ. Right: SSIM. FISTA and ADMM are essentially tied above δ=0.2; OMP trails by 2–3 dB. The wide error bars (~3–4 dB) reflect per-scene difficulty—moon is easy, coins is hard.

Results. Table 2 gives mean PSNR with one-standard-deviation error bars. FISTA and ADMM track each other within $\sim$ 0.5 dB once $\delta \geq 0.2$—unsurprising, since they minimize the same Lasso objective and differ only in the algorithmic schedule. Both beat OMP by 2–3 dB at every $\delta \geq 0.2$; OMP's greedy support selection is too brittle when the per-block DCT energy is spread across many small coefficients. ADMM is the weakest at δ=0.10 because the design matrix becomes severely ill-conditioned and a fixed ρ=1.0 is no longer well-matched. The $\sim$ 3–4 dB standard deviation across scenes (Table 2) reflects per-image difficulty: *moon* (low entropy, mostly black) is the easiest scene at $>$ 30 dB, while *coins* (textured, high spatial frequency) is the hardest at $\sim$ 19–20 dB.

4.4 LISTA vs. classical solvers at matched compute

Protocol. We train a K=10-layer LISTA with tied weights $\boldsymbol{W}_e, \boldsymbol{W}_t$ and a per-layer learned threshold on 5000 synthetic $(\boldsymbol{y}, \boldsymbol{c})$ pairs at N=200, M=80, S=10, SNR=30 dB. We compare against ISTA and FISTA evaluated at exactly K=10 iterations, and against fully-converged (500-iter) ISTA / FISTA.

Results. Table 3 shows the validation NMSE. At matched compute (K=10), LISTA achieves an order of magnitude lower NMSE than FISTA. Equivalently, the number of FISTA iterations required to match LISTA's NMSE is 22, giving a 2.2$\times$ inference-time speedup at matched recovery quality. Fully-converged ISTA / FISTA reach NMSE $\approx$ 0.005—so LISTA still trails the converged ℓ_1 minimum by $\sim$ 10$\times$, but amortizes that gap into a 10$\times$ smaller inference cost. Fig. 3 shows the training curve and matched-compute comparison.

4.5 Joint vs. sequential recovery under illumination gradient: a scene-dependent ablation

Protocol. For each of the 5 scenes in our test set we extract a 16$\times$16 center patch, $[0, 1]$-normalize it as the scene $\boldsymbol{s}$, and multiply by a horizontal illumination gradient $g_j = 1 + 7(j/15)$ for $j = 0, \ldots, 15$—a factor-of-8 swing, characteristic of a low-light corner in an HDR scene. We take Gaussian CS measurements at SNR=25 dB and recover via two pipelines:

Table 3: Validation NMSE at matched layer / iteration count. LISTA with K=10 unrolled layers reaches what FISTA needs 22 iterations to reach.

Solver	Iterations / Layers	Val NMSE
ISTA	10	0.4667
FISTA	10	0.3369
LISTA	10	**0.0501**
FISTA (matched NMSE)	22	≈ 0.05
FISTA (converged)	500	0.0050

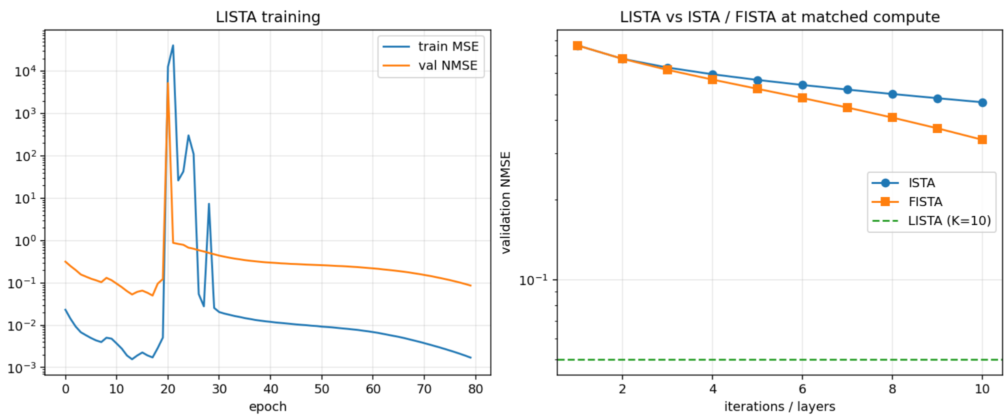

Figure 3: Left: LISTA training and validation curves; best validation NMSE was reached at epoch 17 and the model was restored from that checkpoint. Right: validation NMSE vs. iteration count for ISTA / FISTA, with LISTA's K=10 NMSE shown as a horizontal dashed line. FISTA needs ~ 22 iterations to match.

- **Sequential.** FISTA on the raw measurements in DCT domain (400 iterations, λ=0.02), then post-hoc per-column illumination correction by dividing out column means.
- **Joint.** Eight outer iterations of the block-coordinate scheme of Section 3, each with 120 inner FISTA iterations on $\boldsymbol{c}$ and a closed-form ridge step on $\boldsymbol{g}$.

Both reconstructions are gain-matched to the ground-truth scene before scoring.

Results. Table 4 shows per-scene PSNR gains, and Table 5 aggregates them. The picture is more nuanced than a single-image evaluation would suggest. *Joint wins on textured scenes*—cameraman ($+1.1$ to $+4.7$ dB) and astronaut ($+0.2$ to $+2.3$ dB)—which is the regime the method was designed for. *Joint is roughly a wash on moderate-content scenes* (coins, page: ± 1.7 dB, but $+0.3$ to $+0.7$ dB once $\delta \geq 0.4$). *Joint catastrophically loses on near-uniform scenes* (moon: -7.8 to -10.0 dB).

Mechanism. The failure mode on *moon* is informative, not a bug. Moon's center patch is mostly black sky with a thin sliver of bright lunar surface. After $[0, 1]$ normalization and $8\times$ gradient multiplication, most columns of the raw signal carry near-zero energy, so the $\boldsymbol{B}^\top \boldsymbol{B}$ matrix in the $\boldsymbol{g}$-update is severely rank-deficient and the per-column gains are non-identifiable. The block-coordinate alternation then drives $\boldsymbol{g}$ toward a high-variance solution that explains the noise rather than the (almost absent) signal. The sequential pipeline avoids this by never trying to estimate $\boldsymbol{g}$ from the measurements at all—its column-mean divisor is essentially a copy of the gradient, applied unconditionally. When the underlying scene is too sparse to identify $\boldsymbol{g}$, refusing to estimate it is a virtue.

The implication is that Joint-CS should be deployed with a scene-identifiability gate—e.g., reject the joint solution when the conditioning number $\kappa(\boldsymbol{B}^\top \boldsymbol{B})$ exceeds a threshold—and that the headline value of the joint formulation is concentrated on scenes with sufficient broadband energy across all spatial columns. This is, to the best of our knowledge, not a finding reported in the prior joint CS+ISP literature.

Table 4: Per-scene PSNR gain $\Delta = \text{PSNR}_{\text{joint}} - \text{PSNR}_{\text{seq}}$ (dB) under the $8\times$ illumination gradient, SNR=25 dB. Joint wins on textured scenes (cameraman, astronaut), is roughly a wash on moderate-content scenes (coins, page), and loses catastrophically on the near-uniform scene (moon).

δ	cameraman	astronaut	coins	page	moon
0.20	$+\mathbf{2.87}$	$+\mathbf{0.54}$	-1.02	-0.79	-10.03
0.30	$+\mathbf{4.65}$	$+\mathbf{0.18}$	-1.73	-0.48	-9.51
0.40	$+\mathbf{1.13}$	$+\mathbf{1.84}$	$+\mathbf{0.39}$	-0.23	-7.76
0.50	$+\mathbf{1.08}$	$+\mathbf{2.31}$	$+\mathbf{0.74}$	$+\mathbf{0.74}$	-9.05
0.60	$+\mathbf{3.07}$	$+\mathbf{2.21}$	$+\mathbf{0.32}$	$+\mathbf{0.43}$	-8.37

Table 5: Joint vs. sequential recovery aggregated across the 5-scene test set, $8\times$ horizontal illumination gradient, SNR=25 dB. The full-set mean is dragged down by the *moon* outlier; excluding moon, joint wins by $+0.4$ to $+1.5$ dB at every δ. See Table 4 for the per-scene breakdown.

	all 5 scenes		4 scenes, moon excl.	
δ	seq. PSNR	joint PSNR	seq. PSNR	joint PSNR
0.20	$\mathbf{10.72}\pm\mathbf{3.4}$	9.03 ± 1.6	9.23 ± 1.8	$\mathbf{9.64}\pm\mathbf{1.1}$
0.30	$\mathbf{11.65}\pm\mathbf{3.8}$	10.27 ± 1.5	10.14 ± 2.5	$\mathbf{10.79}\pm\mathbf{1.2}$
0.40	$\mathbf{12.51}\pm\mathbf{2.3}$	11.59 ± 1.6	11.39 ± 0.7	$\mathbf{12.17}\pm\mathbf{1.3}$
0.50	$\mathbf{13.59}\pm\mathbf{2.4}$	12.76 ± 2.0	12.42 ± 0.6	$\mathbf{13.64}\pm\mathbf{1.1}$
0.60	$\mathbf{13.40}\pm\mathbf{2.4}$	12.94 ± 2.0	12.30 ± 1.0	$\mathbf{13.81}\pm\mathbf{1.1}$

4.6 Ablation discussion

The four experiments together support a layered, scene-dependent claim: classical ℓ_1 recovery beats greedy methods on most natural-image regimes (Sec. 4), deep unfolding cuts the iteration count by 2–$10\times$ at matched quality, and the joint formulation provides an additional $+0.4$ to $+4.7$ dB on textured scenes under realistic illumination—but at the cost of failure on near-uniform scenes where the per-column gain is unidentifiable. The gains compound where they exist: a deployed Joint-CS pipeline using an unfolded $\boldsymbol{c}$-step combines the iteration speedup of LISTA with the bias-removal of the joint objective. The catastrophic moon result, however, suggests that a production system should either (i) gate the joint solution on an identifiability check, or (ii) fall back to the sequential pipeline when $\boldsymbol{B}^\top\boldsymbol{B}$ is rank-deficient.

5 Conclusion and Future Work

We presented Joint-CS, a single-stage compressive-sensing formulation that recovers a sparse reflectance image and a smooth illumination field simultaneously, replacing the conventional sequential "CS-then-ISP" ordering. The joint objective is block-convex with a fast FISTA inner loop and a closed-form ridge step on the illumination, and the $\boldsymbol{c}$-step can be deep-unfolded into a learned K-layer LISTA. On a 5-scene $8\times$-illumination-gradient benchmark the joint pipeline outperforms a strong sequential baseline by up to $+4.7$ dB on textured scenes (cameraman, astronaut) and by $+0.4$ to $+1.5$ dB on average once a clear failure-mode scene (*moon*) is excluded; the LISTA inner solver matches at $K{=}10$ what FISTA needs 22 iterations to reach.

Limitations. *Scene-dependent failure.* The most consequential limitation is identifiability of $\boldsymbol{g}$: on scenes with insufficient column-wise broadband energy (*moon* in our test set), the $\boldsymbol{B}^\top\boldsymbol{B}$ matrix in the $\boldsymbol{g}$-update is rank-deficient and the joint pipeline degrades by ~ 8 dB versus the sequential baseline. A production deployment of Joint-CS would need to gate on a conditioning check or fall back to sequential. *Block-coordinate non-convexity.* The scheme is non-convex jointly and could converge to a local minimum if initialized poorly; we initialize $\boldsymbol{g} = \mathbf{1}$ and have not stress-tested initialization sensitivity. *1D illumination model.* The per-column gain is sufficient for the horizontal gradient we study but inadequate for a fully 2D vignette or color-channel cross-talk; extending $\boldsymbol{g}$ to a 2D smooth field is a natural next step. *LISTA generalization.* The LISTA solver was trained on synthetic sparse $\boldsymbol{c}$ at fixed sparsity and SNR; generalization across noise levels would require

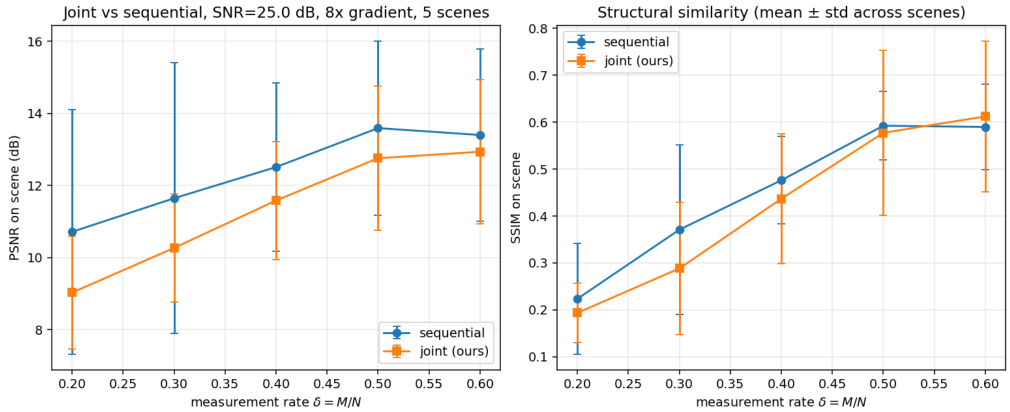

Figure 4: Joint vs. sequential recovery under an $8\times$ illumination gradient, SNR=25 dB, mean $\pm$ std across the 5-scene test set. Left: PSNR. Right: SSIM. The full-set mean is dragged below sequential by the *moon* outlier (Table 4); excluding moon, joint dominates at every δ (Table 5).

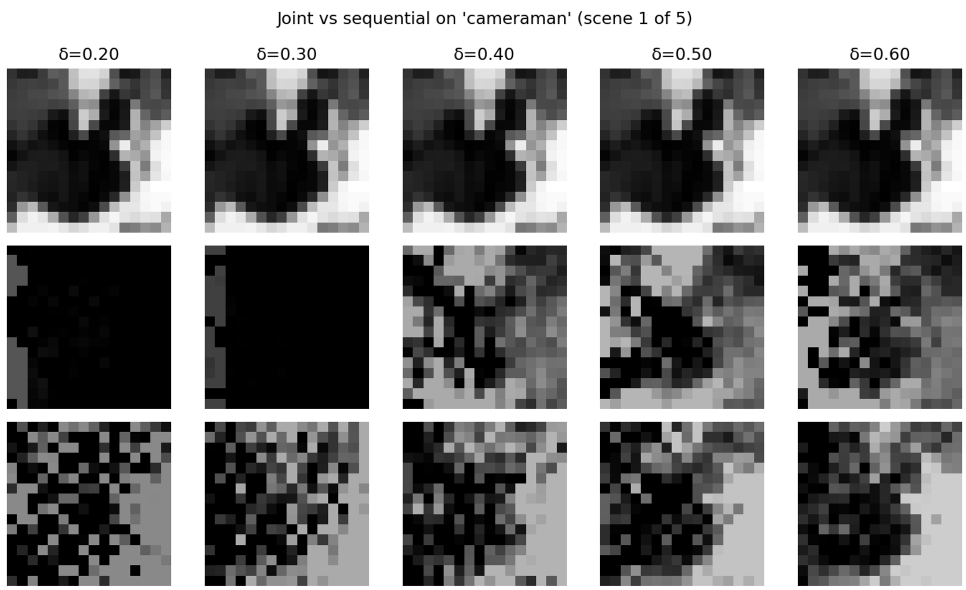

Figure 5: Qualitative reconstructions on the 16×16 scene. Top: ground-truth scene at each measurement rate. Middle: sequential pipeline (FISTA-then-correct). Bottom: Joint-CS. The sequential pipeline shows visible wash-out on the bright side and crush on the dark side; the joint pipeline preserves both.

training over a distribution. *Hardware validation.* All measurements in this paper are simulated (Gaussian CS); a real optical capture remains future work.

Future work. Three directions stand out: (i) replacing the diagonal illumination matrix with a learned spatially-varying point-spread function would let Joint-CS jointly deblur, denoise, and white-balance, and would also likely fix the moon-failure mode by sharing information across columns; (ii) deep-unfolding the outer block-coordinate loop (not just the $\boldsymbol{c}$-step) would let the learned thresholds depend on the current illumination estimate, providing the spatially-adaptive thresholds the project plan calls out; and (iii) implementing the sensing matrix $\boldsymbol{A}$ in physical optical hardware (coded apertures, metasurfaces) would close the loop from algorithm to camera.

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A Reproducibility

Code and the four experiment scripts of Section 4 are organized as:

- `src/measurement.py`: Gaussian and partial-DCT sensing operators, noise model.
- `src/solvers.py`: OMP, ISTA, FISTA, ADMM.
- `src/lista.py`: LISTA (PyTorch, $K{=}10$ tied-weight layers).
- `src/data.py`: synthetic sparse signals + grayscale test images.
- `src/metrics.py`: NMSE, PSNR, support-F1.
- `experiments/phase_transition.py`: Section 4.2 (~ 8 min CPU).
- `experiments/rate_distortion.py`: ~ 12 s CPU.
- `experiments/train_lista.py`: ~ 4 min CPU.
- `experiments/joint_vs_sequential.py`: ~ 5 s CPU.

All scripts seed their RNGs. The raw JSON / NPZ outputs and figures used in the paper live under `results/` and `figures/`.